

Making BO stable on the enzyme benchmark

Landscape screens, four campaigns, a rejected selection, and the surrogate study now running

11 August 2026

1. Status

Not concluded. The design-space route was explored, produced a statistically clean answer, and that answer was **rejected on physical realism** (§5). The work has since moved to the surrogate, where the instability originates, and that study is running. This report separates what was *decided*, what was *measured*, and what is still open — several of the measurements are sound while the conclusions first drawn from them were not.

The problem being fixed: **meta** fits **3–6× more BO iterations** into the same detector-call budget than **from_scratch** or **closest**, and the benchmark is only meaningful if that converts into a lower objective. On the configured box it did the opposite — **meta** reached ~0.094 in 22k calls where **continue** needed ~82k, then plateaued at 117k and finished **worst** (verified 0.0842 against **continue**’s 0.0621 from a third as many designs).

2. Decisions taken by the user, and the motivation for each

These set the direction; everything in §3–§8 is subordinate to them.

#	Decision	Motivation
D1	The enzyme system is off limits — noise, duration, read-outs, kinetics, the T_melting prior	the simulated population must keep genuine variability; tuning the physics makes a different, easier problem rather than a better benchmark
D2	The search box is the priority — “ not too narrow, not too wide ”; if stability requires a tight box, tune something else	a box narrowed until it works is a benchmark with the answer written into it
D3	Kernel: the normalised symmetric one	it was the most stable in earlier work. Confirmed here at k=30: 13/16 seeds beat the null, against sorting’s 10/16 and the bare group average’s 6/16
D4	loss_precision stays 1.0e-2; the budget pays for it (524288, iteration_limit = budget)	the criterion’s slack must be smaller than the differences BO must resolve. At 2.0e-2 the slack (0.02) exceeded the ~0.012 gap between a good design and the best one, so the optimum is invisible however many iterations are spent
D5	Horizon 15 → 30 iterations	the range the neural campaign actually spans: from_scratch completes 14–24 designs a run
D6	Judge on stability, not on beating random search	the object of study is the training strategies; “BO out-runs a shotgun” is not the goal
D7	[42, 69] rejected as unrealistic	it leaves 3 °C below the lowest melting point and 11 °C above the highest. No experimenter knowing only T_m in [45, 58] would choose that, so it encodes knowledge of where the answer lies

#	Decision	Motivation
D8	Attack the flat GP instead — narrow the lengthscale prior	the failure looks like <i>uninformative initial designs</i> → <i>nearly constant objective</i> → <i>flat GP</i> → <i>EI degenerates toward random</i> . Fixing the optimiser’s prior does not narrow the problem
D9	Observation noise from the regressor’s own gap: SD $\text{abs}(\text{val} - \text{train})$ for the proxy and $\text{abs}(\text{val} - \text{train})/2$ for the neural objective, i.e. $\alpha = \text{abs}(\text{val} - \text{train})^2$ and $\text{abs}(\text{val} - \text{train})^{2/4}$	the reported value is $(\text{train} + \text{val})/2$, so its uncertainty is the spread of the two numbers averaged — not the sampling error of either, which the convergence test has already bounded

D6 corrected a mistake of mine. I had made “beats the null” the headline statistic, and that is what drove the rejection of the fraction-narrowed arms in §5. Re-judged on BO’s own curve those arms still fail — but because $\text{best@15} / \text{best@30} = 1.000$, i.e. BO stops improving after iteration 15, not because they lose to random. They are stable in the uninteresting sense: every design is equally good.

2b. Decisions taken by me

The table above is the direction. Everything below is mine — the experimental design that turned those instructions into runs. They are listed separately because they are the choices a reader should scrutinise: all four corrections in §7 arose here, and one was overturned outright.

#	Decision	Motivation	Outcome
A1	The statistic: per-seed payoff over the doubling, and a ≥ 1.15 threshold	the unit of replication must be the seed; the project’s audit had condemned a ratio of medians in the previous box decision	corrected twice — I nonetheless computed a ratio of medians (§7.1), and D6 later demoted the null comparison entirely
A2	Screen 5 box widths, then 11 including asymmetric centres, before any campaign	~336 evaluations a variant against a campaign’s ~1500; cheap enough to map the trade first	sound; the screens’ <i>rankings</i> then mispredicted the campaign twice
A3	Test the fraction at $[0.2, 0.7]$, from the measured profile	the loss profile is flat over 0.3-0.6, so the tails looked like dead volume	rejected by campaign — it flattens the curve (§5)
A4	16 seeds an arm, seeds 0-7 selecting and 8-15 confirming, criterion fixed before launch	the previous box decision was criticised as circular: chosen and reported on the same runs	held; [42, 69] was the only arm whose confirmation half beat its selection half
A5	Select [42, 69] on payoff, spread and worst seed	best on every point estimate available at the time	overturned by D7 — I optimised a population-average statistic and let it pick a box that encodes where the answer lies
A6	A 32-seed two-arm rerun of the one comparison that mattered	nine arms cost a nine-way multiplicity price; the decisive claim deserves its own test at double the power	sound; it is what showed the head-to-head is unresolved

#	Decision	Motivation	Outcome
A7	Bootstrap CIs, Holm across arms, two-sided tests throughout	16 seeds cannot deliver significance against effects this size, so the intervals had to be shown rather than a p-value quoted	held
A8	Revert the box to [25, 80]; stop and archive the box campaigns rather than delete them	a rejected setting must not sit in the config looking settled, and superseded runs are still evidence	held
A9	Expose the GP priors as CLI flags; pick the four prior widths under test	D8 needed them and the driver hardcoded them; the widths bracket the fitted lengthscale (0.24-0.29) and the excursions (0.59)	in progress

3. What was measured without an optimiser in the loop

Landscape screens score random designs directly — 192 per variant in the first round, 384 in the second — so nothing here can be an artefact of the optimiser.

The old box starves the GP of initial signal. Expected number of informative designs among the five random initial draws: **2.9 of 5** at [25, 80], 4.2 at [35, 68], 4.9 at [42, 62]. This is the diagnosis the whole exercise started from, and it is what D8 now attacks from the surrogate side rather than by moving the box.

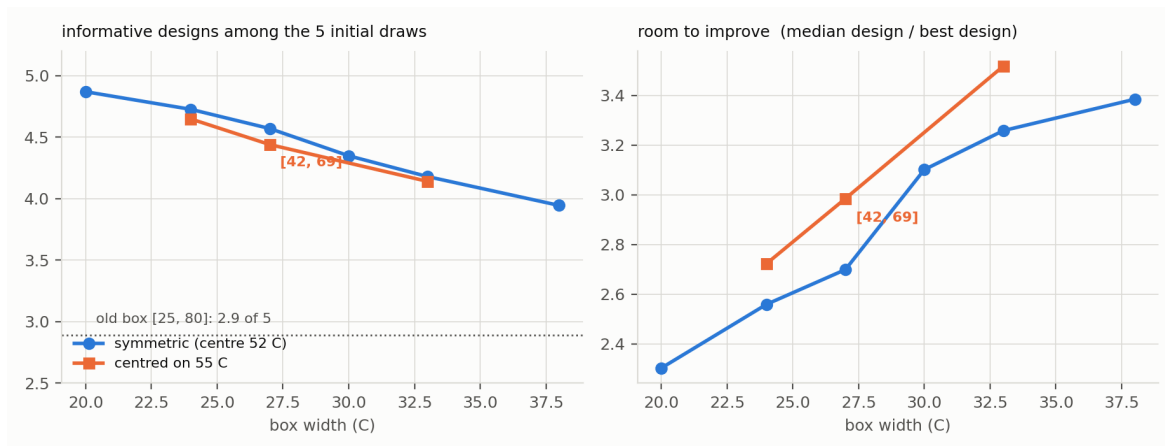


Figure 1: The width trade. Left: informative designs among the five initial draws. Right: room between the median design and the best one. Signal falls monotonically with width while room grows — no width maximises both. The upper series is the asymmetric family later rejected under D7.

The enzyme fraction has a broad flat optimum at 0.3–0.6, which motivated narrowing it — a change the campaign then reversed (§5).

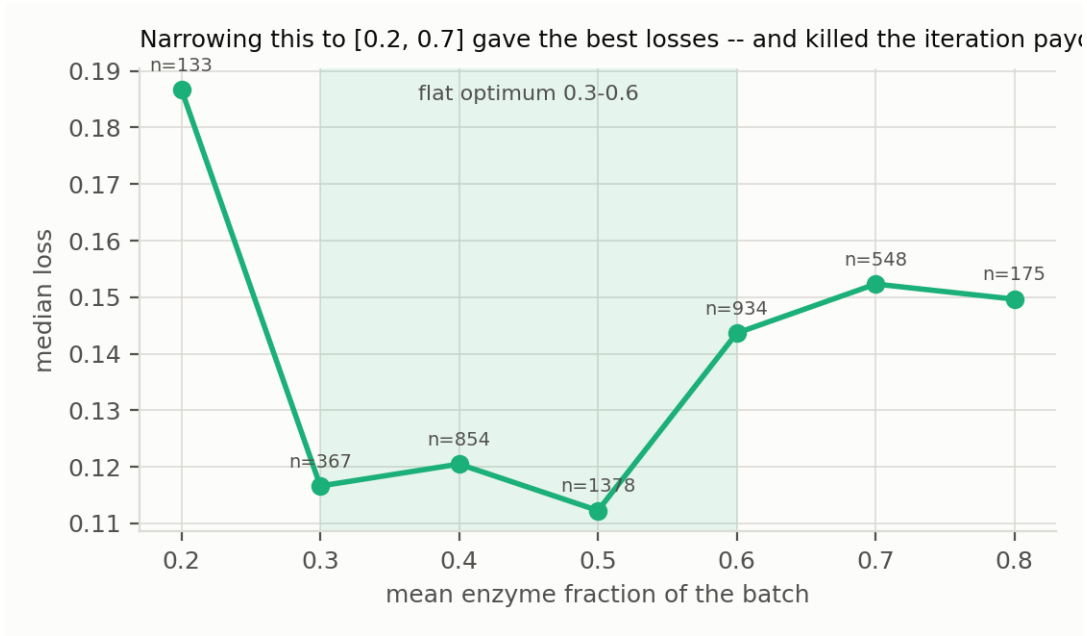


Figure 2: The fraction profile, from the same screens at no extra cost. Its flatness is exactly why narrowing to [0.2, 0.7] improved the loss and destroyed the iteration payoff.

4. The campaigns

All on the GBDT proxy (~ 0.7 s a design against 27–137 s; Spearman 0.940 against the neural loss), 60 iterations, no failures anywhere:

campaigns $\times$ seeds	question	state
output 0/box (16+16)	which design space converts iterations	done
output 2/box (32+32)	selected box vs incumbent, double the seeds	done
output 3/real (16+16)	boxes with a defensible lower bound	stopped under D7/D8
output 4/lengthscale, 16 each	the lengthscale prior at the incumbent box	running

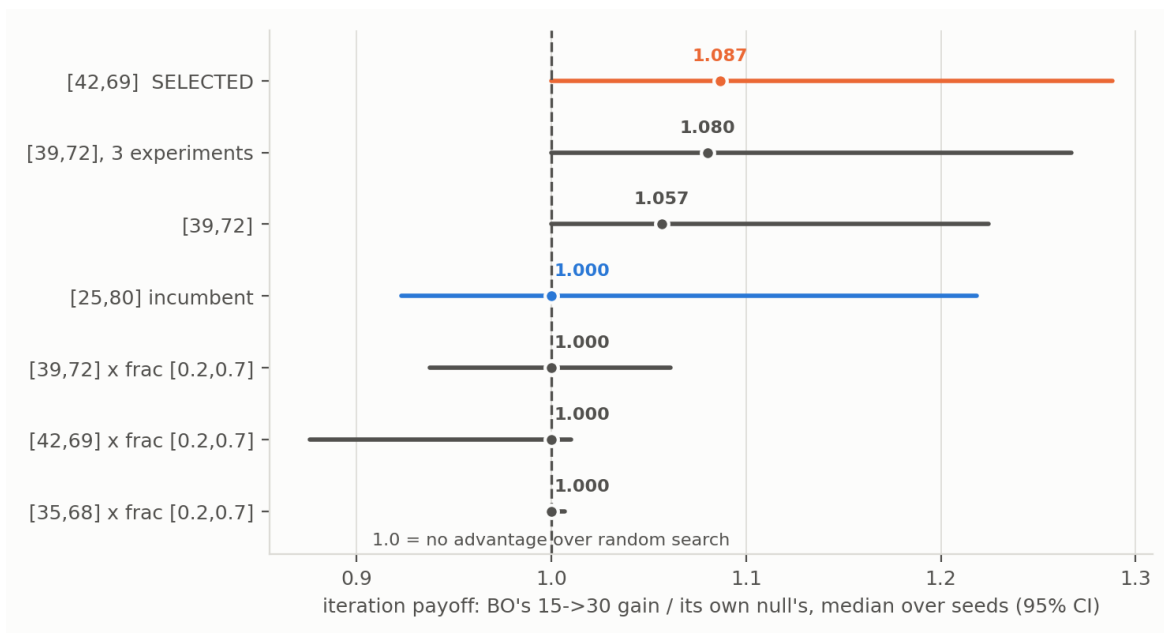


Figure 3: Every arm’s payoff against its own null with bootstrap CIs. Read as a measurement of the design space, not as the selection criterion — D6 replaced that.

5. What was rejected, and why

[42, 69] — **the statistically strongest result — on realism (D7)**. At 32 seeds it gave payoff 1.104, CI [1.006, 1.181], $p = 0.004$, against the incumbent’s 1.033 at $p = 0.191$: a clean result on the pre-registered criterion, and still unusable. Measured from `gibbs_fraction` at the prior’s extremes:

enzyme	transition span (90% → 10% folded)
$T_m = 45, \Delta H$ mid	41.7 – 47.9 °C
$T_m = 45, \Delta H$ low	25.9 – 49.3 °C
$T_m = 58, \Delta H$ mid	54.4 – 61.1 °C

A bottom-of-prior enzyme’s informative probes start at 41.7 °C and reach to 25.9 °C for broad-transition ones, so the box clips precisely what the hardest enzymes need; nothing above ~61 °C is informative for any enzyme, so 8 °C of its range is dead by construction. **temperature_bounds** is back at [25, 80].

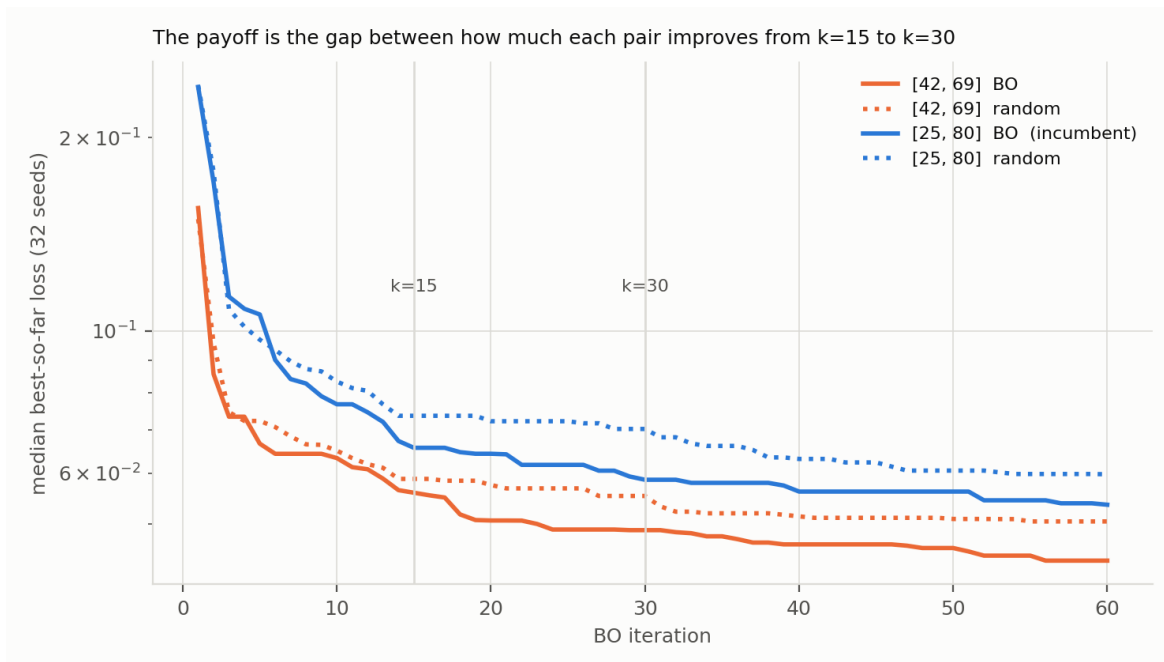


Figure 4: The 32-seed comparison behind the rejected selection. The measurement is sound — the orange BO/null pair separates between the marked horizons while the blue pair stays parallel — but the box it selects is not admissible.

Narrowing the enzyme fraction to $[0.2, 0.7]$: best absolute losses anywhere (0.0404, 1.15 °C) and the tightest seed spread (IQR 0.0038) — and $\text{best@15}/\text{best@30} = 1.000$. Removing the $f \rightarrow 0$ and $f \rightarrow 1$ corners leaves every design nearly as good as every other. A milder $[0.1, 0.9]$ is queued.

Three experiments instead of four: payoff comparable, absolute loss far worse (0.0687 against 0.0478). A smaller batch makes the problem harder, not more learnable.

Widening the transition via ΔH — rejected before it cost a campaign: halving the unfolding enthalpy drops peak folded fraction to 0.665 and worsens the loss $\sim 1.7\times$. A different objective, and a violation of D1.

6. The line of attack now running

Under D8 the target is the surrogate, at the unchanged box. Two defects were measured in it.

The lengthscale is unstable exactly when it matters. Median fitted l by iteration: 0.59 at 6, 0.25 at 7, 0.36 at 8, 0.58 at 10 — settling to ~ 0.25 only after iteration 15. Those early fits drive the first GP proposals, each under a materially different smoothness assumption. The prior $\log l$ in $[-6, 4]$ permits l from 0.0025 (every point independent) to 54.6 (the objective is constant): both degenerate, both inside the prior. Four narrower priors are under test.

The GP was forced to interpolate. Alpha was the sampling SEM — $\text{sem}^2 = 6.5\text{e-}06$ against a fitted prior variance of 0.008, a ratio of **1:1200**. Step-like structure then has to be explained by shortening the lengthscale. Under D9 the SD is now the generalisation gap, $|\text{val} - \text{train}| \sim 0.050$, so $\alpha \sim 0.0025$ — **31% of the prior variance**, and the surrogate can attribute roughness to noise again. `BayesianOptimizer.append` takes an SD and `_fit` squares it, so D9’s $\alpha = |\text{val} - \text{train}|^2/4$ on the neural side is obtained by passing $0.5 \cdot |\text{val} - \text{train}|$.

For calibration, the prior *scale* was checked and is not at fault: fitted at $0.99\times$ the data variance from iteration 6 onward, never at a bound. Bounding it from the objective’s known support (a normalised MSE — measured

min 0.032, p95 0.325, no-information level 1/3) is reasonable hygiene and is wired as a flag, but it is not the defect.

7. Corrections made during the work

All recorded in `docs/enzyme-retune-log.md`:

1. **Ratio of medians.** The payoff was first computed as $\text{median}(\text{BO gains})/\text{median}(\text{null gains})$, inflating the headline from 1.087 to 1.207 — the exact defect this project’s audit had named. Rewritten to the per-seed statistic and every number re-derived.
2. **A design count projected from four samples** — read at 4 designs and extrapolated to ~ 8 a run; left running it reached 10 and was heading for ~ 12 . The expensive designs are the early ones.
3. **An attribution to patience, then to the epoch length** — an isolating run refuted both. Runs at different budgets do not evaluate the same designs, so single-design comparisons across them measure nothing.
4. **A prior-variance ratio wrong by a square** — `log_amplitude` is recorded as $0.5 \cdot \log(\text{variance})$, so exponentiating it and comparing against a variance gave a bogus “11.7 $\times$ over-dispersed”; the true figure is 1.08 $\times$.

Two operational catches: `snakemake` silently reused a stale seed from before the retune (its `results.json` existed, and config files are deliberately not declared inputs), which would have contaminated the neural comparison; and a wait-loop whose condition contained its own search string matched itself and span for 25 minutes without launching the campaign behind it.

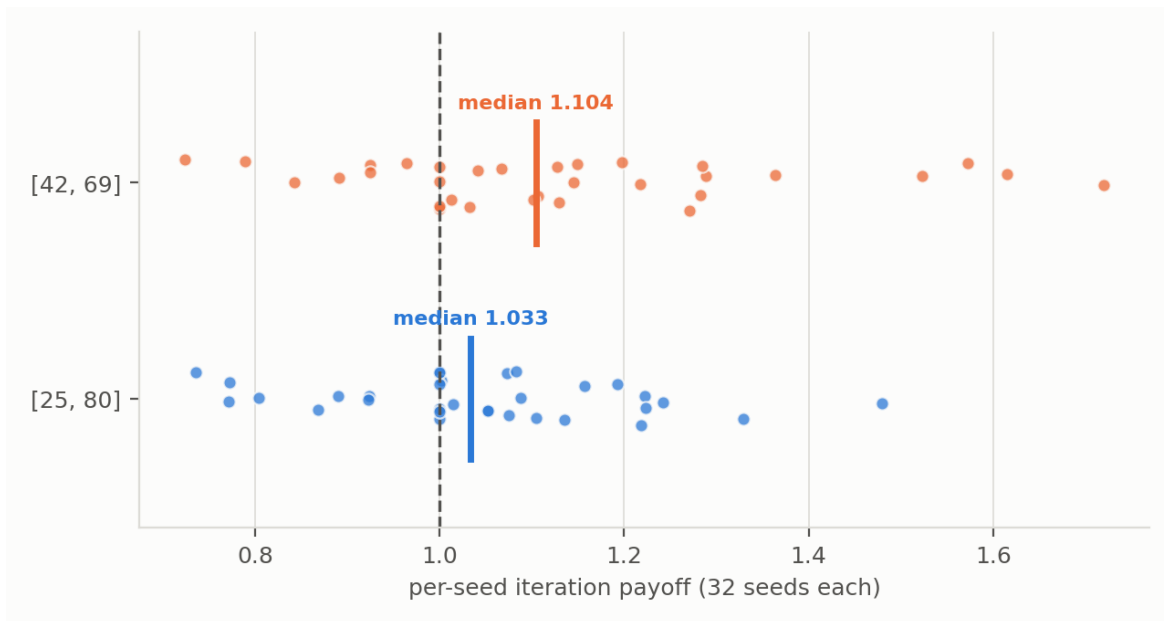


Figure 5: Per-seed payoffs behind the medians of the rejected selection. The overlap is why the head-to-head between boxes was never resolved even at 32 seeds.

8. What is not established

- That any box converts iterations *better than* another. Head to head, [42, 69] vs [25, 80] differ by +0.071, bootstrap CI [-0.050, +0.171], $p = 0.16$. The claim was only ever the two separate one-sample results, and that box is rejected regardless.
- Anything on the **real objective**. Every number here is the GBDT proxy; the neural campaign was stopped when its box was rejected and will be re-run once the surrogate settings settle.

- Whether the new noise model over-smooths. At 31% of the prior variance the GP may now under-fit real structure — the opposite failure. The running arms will show it: if every lengthscale prior yields the same flat curve, the nugget is too large and belongs between the SEM and the gap.

9. Reproduction

```
# landscape screens (no optimiser in the loop)
python scripts/enzyme_landscape.py screen --n-designs 384 --n-events 8192 --n-anchors 16 \
  --set enzyme.temperature_bounds=[25.0,80.0] --output output/screen1/base.json
```

```
# campaigns
./scripts/gbdt_campaign.sh scripts/arms/box.arms          output/box          16 16 60 10
./scripts/gbdt_campaign.sh scripts/arms/box32.arms        output/box32        32 32 60 6
./scripts/gbdt_campaign.sh scripts/arms/lengthscale.arms  output/lengthscale 16 16 60 8
```

Config as it now stands: `temperature_bounds: [25.0, 80.0]` (reverted under D7); `loss_precision: 1.0e-2`, `budget/iteration_limit 524288`, `warmup_epochs: 16`, `patience: 8` (D4); kernel `normalised-invariant-rbf` (D3). Code changes are confined to exposing the GP priors as CLI flags on the proxy driver, and the noise model of D9.